

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

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S A S 3 0 3

Inside the SaaS control plane

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Chalk talk



Getting to know you

By a show of hands ...

1. Are you currently operating an existing SaaS solution today?
2. Are you actively building a SaaS solution today?
3. Are you in the planning or designing phase of building a SaaS solution?
4. Did you have nothing better to do? 😊

SaaS value proposition



Agility



Innovation



Operational
efficiency

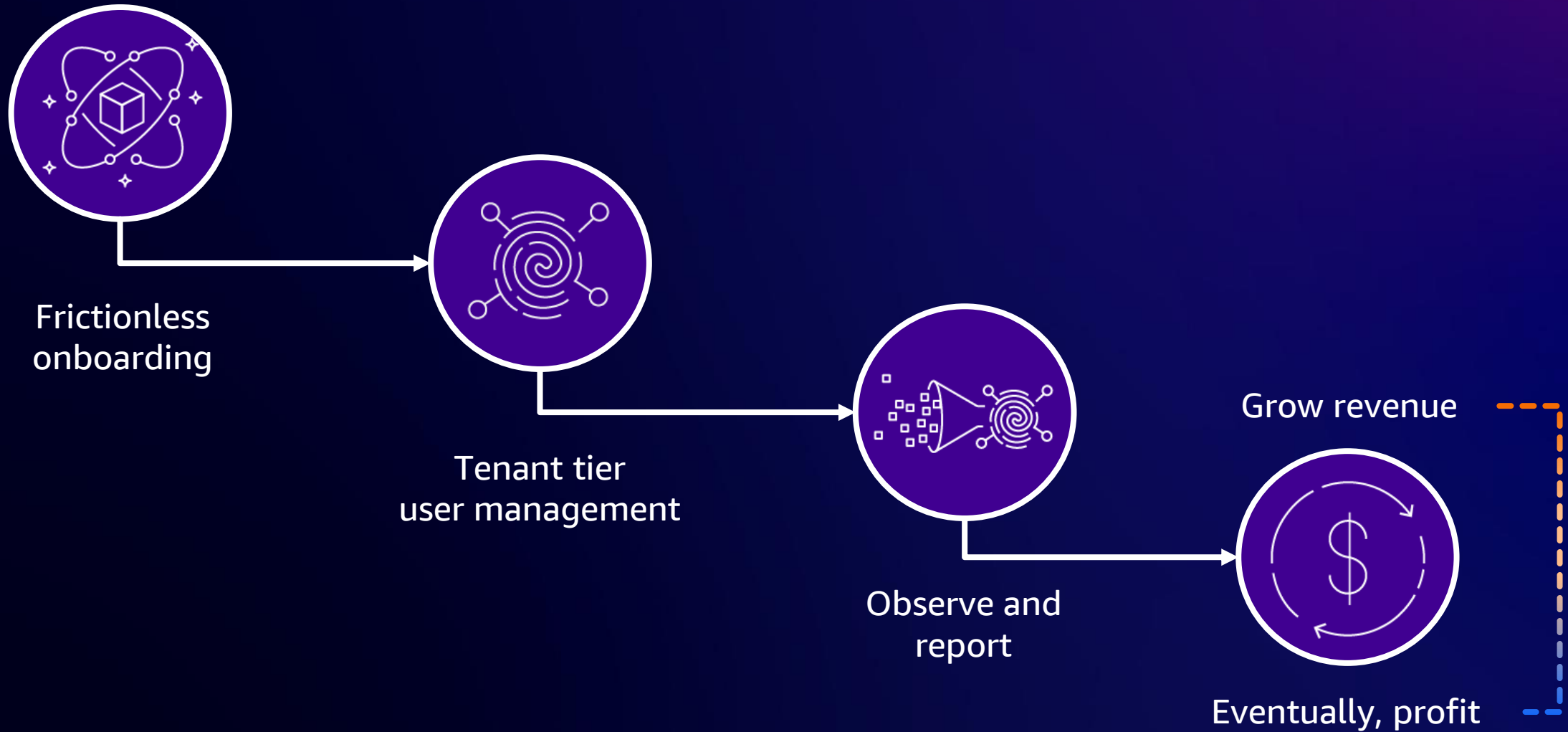


Growth



Reduced cycle
time

Control plane = SaaS growth



SaaS control plane responsibilities



Single pane
of glass



Operations



Orchestration



Observability

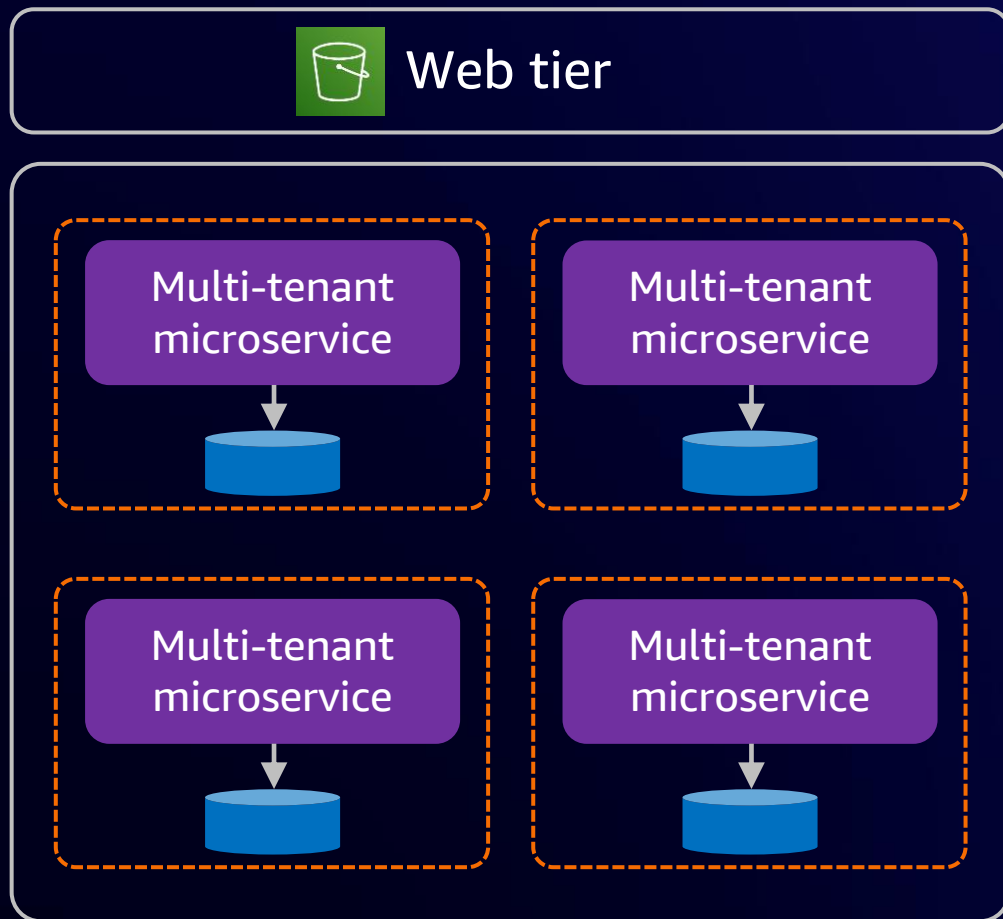


Integration

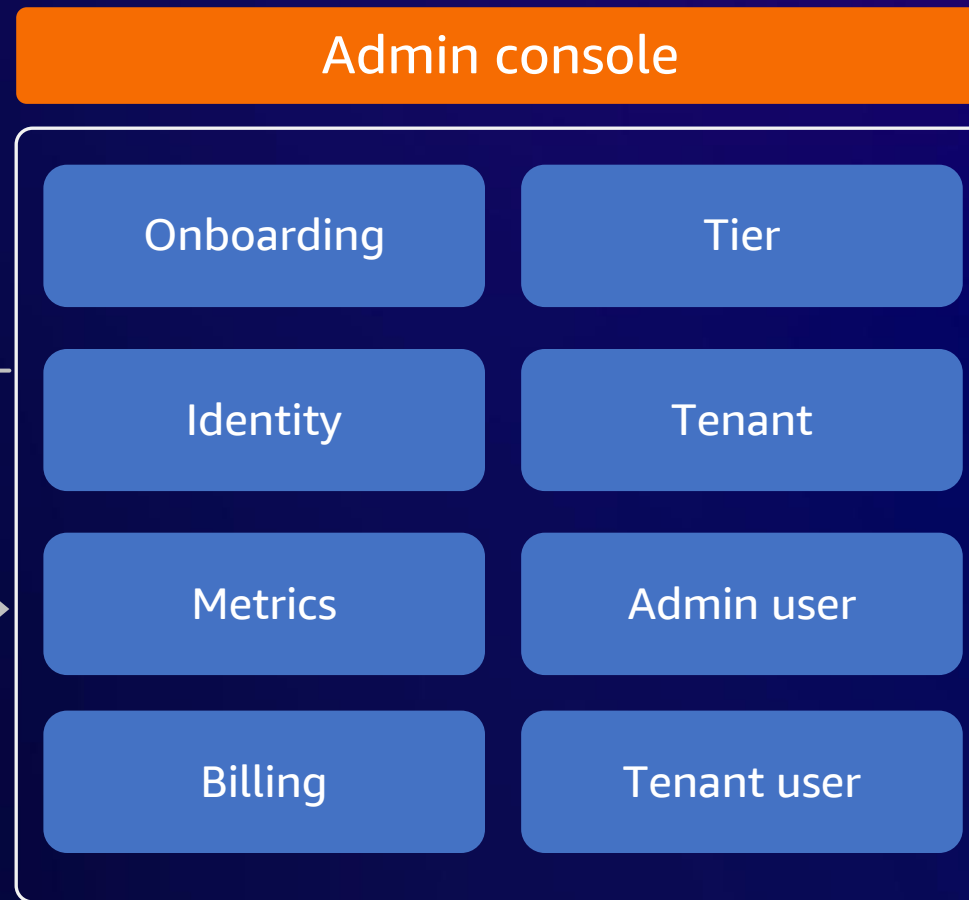
Question: What are some control plane services?

The two halves of SaaS

Application plane



Control plane



Onboarding actions and events

SaaS
admin



Onboarding

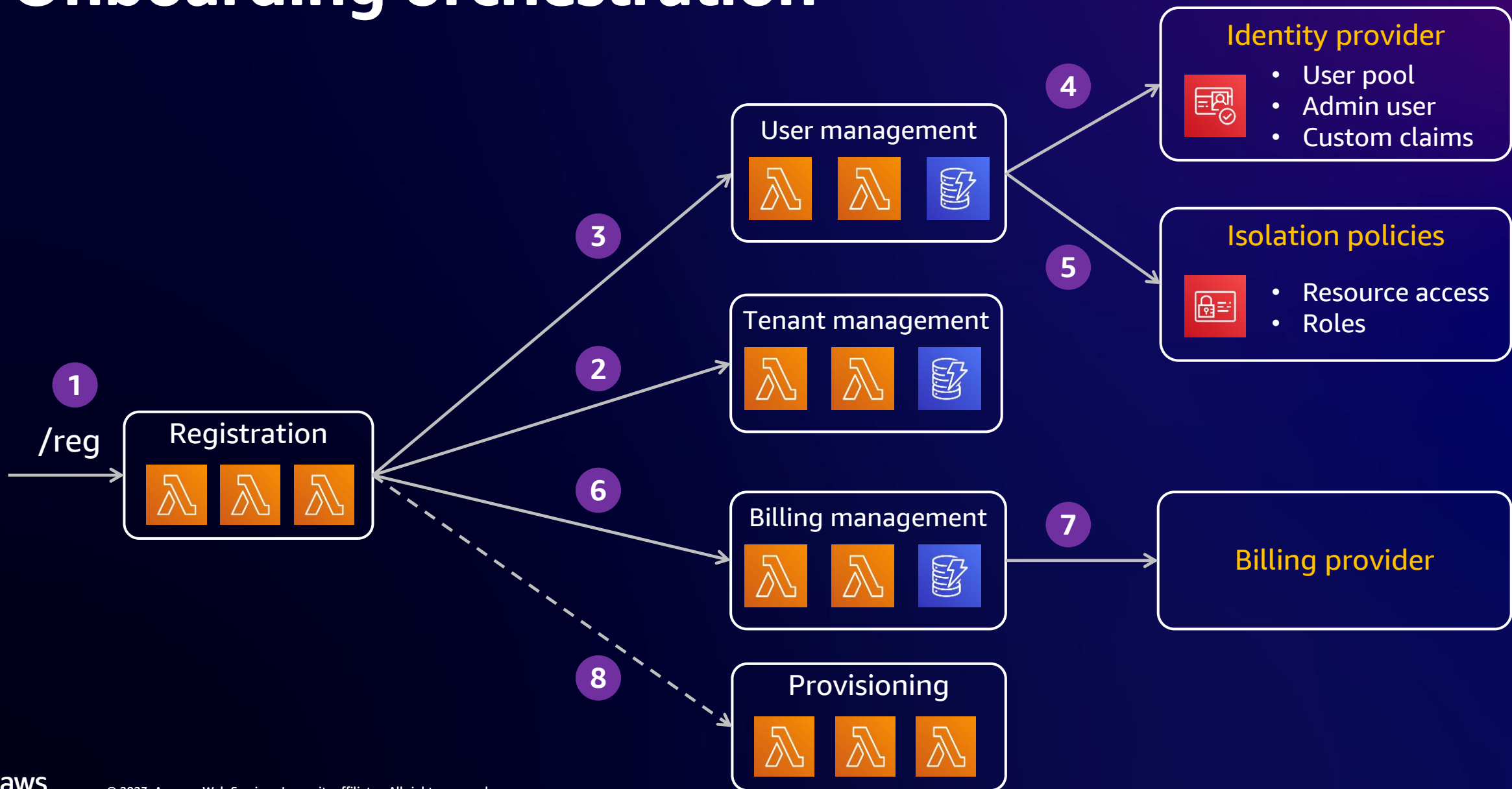
Actions

- 1 Save onboarding request and enqueue for validation
- 2 Validate environment
- 3 Create tenant record
- 4 Provision infrastructure
- 5 Execute deployment pipelines
- 6 Wire up 3rd party billing provider
- 7 Change status of tenant record

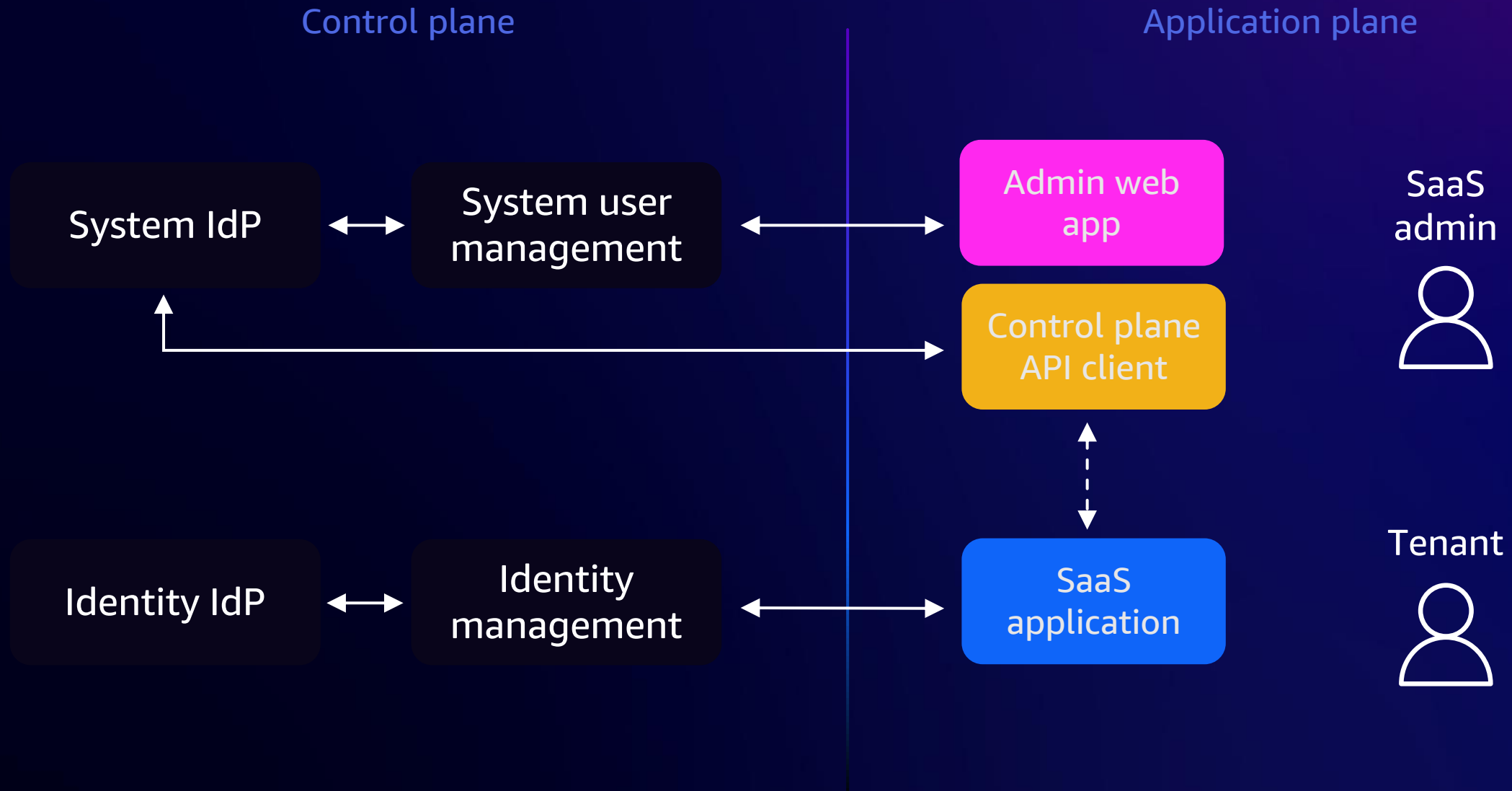
Events

-  Onboarding initiated
-  Onboarding validated
-  Onboarding tenant assigned
-  Onboarding provisioned
-  Onboarding deployed
-  Onboarding billing integrated
-  Onboarding complete

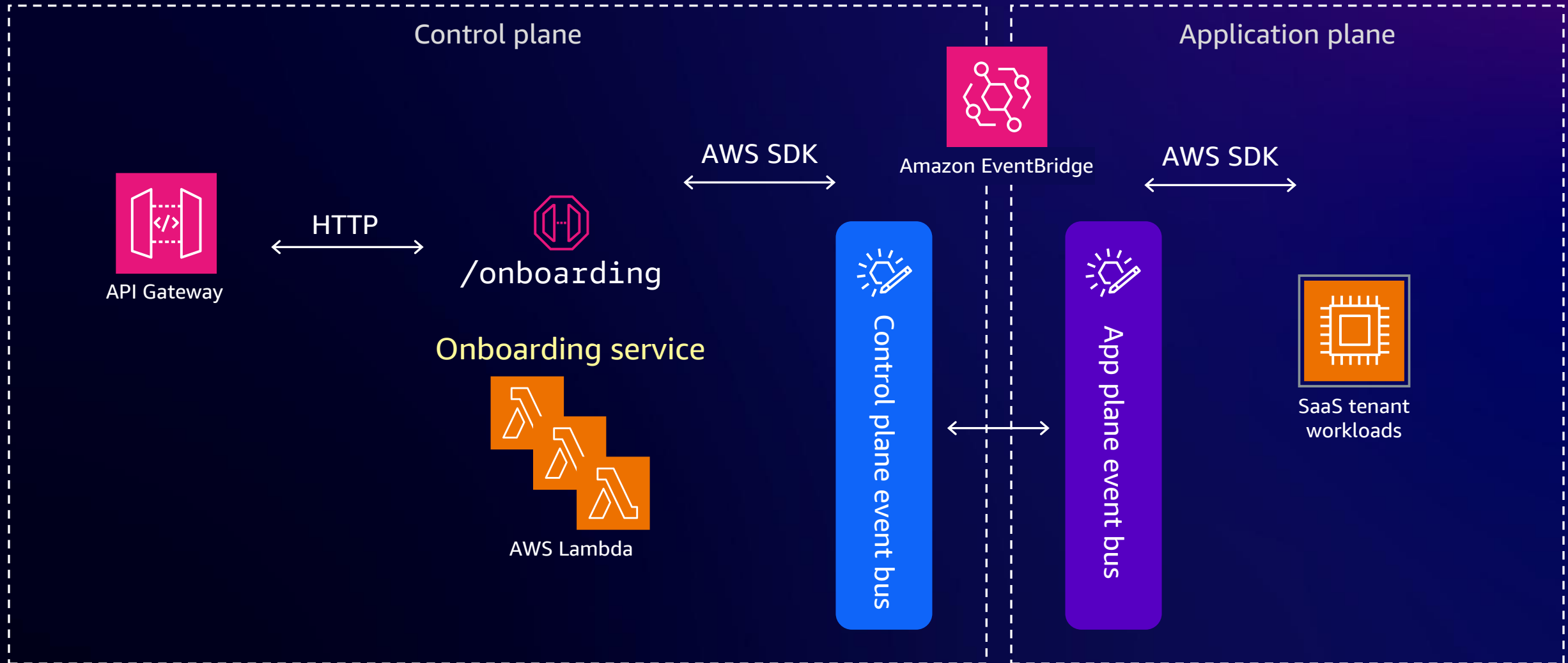
Onboarding orchestration



Identity blurs the lines



Onboarding and event-driven architecture



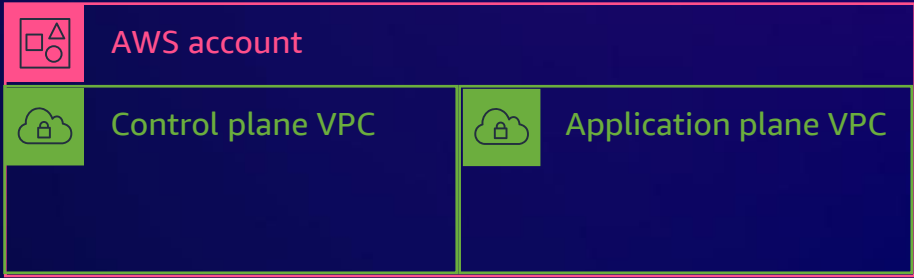
**Question: What are some
control plane, application plane
architecture patterns and
examples?**

Control plane architectures

Multi-account



Multi-VPC



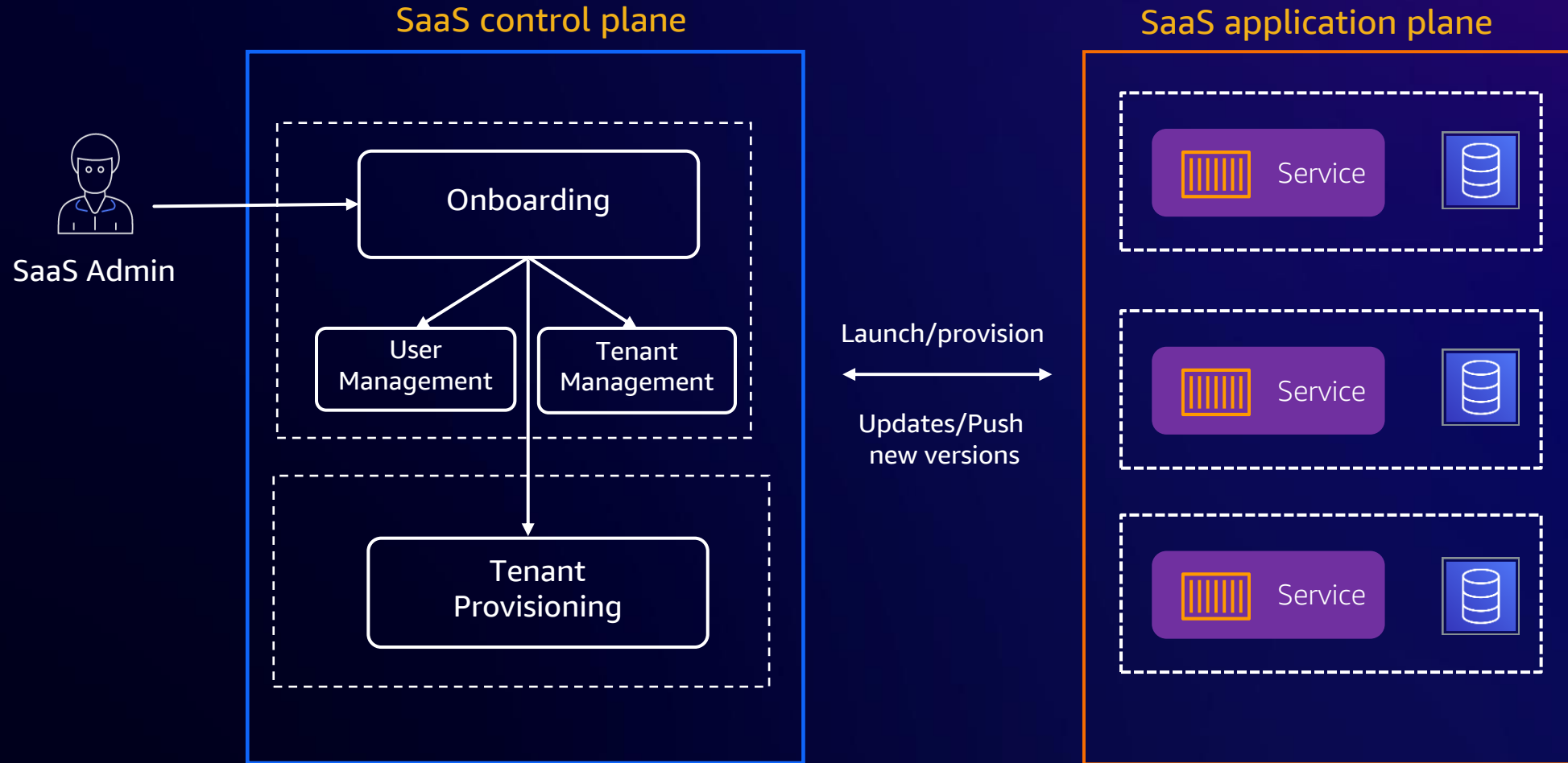
Cross-customer/provider



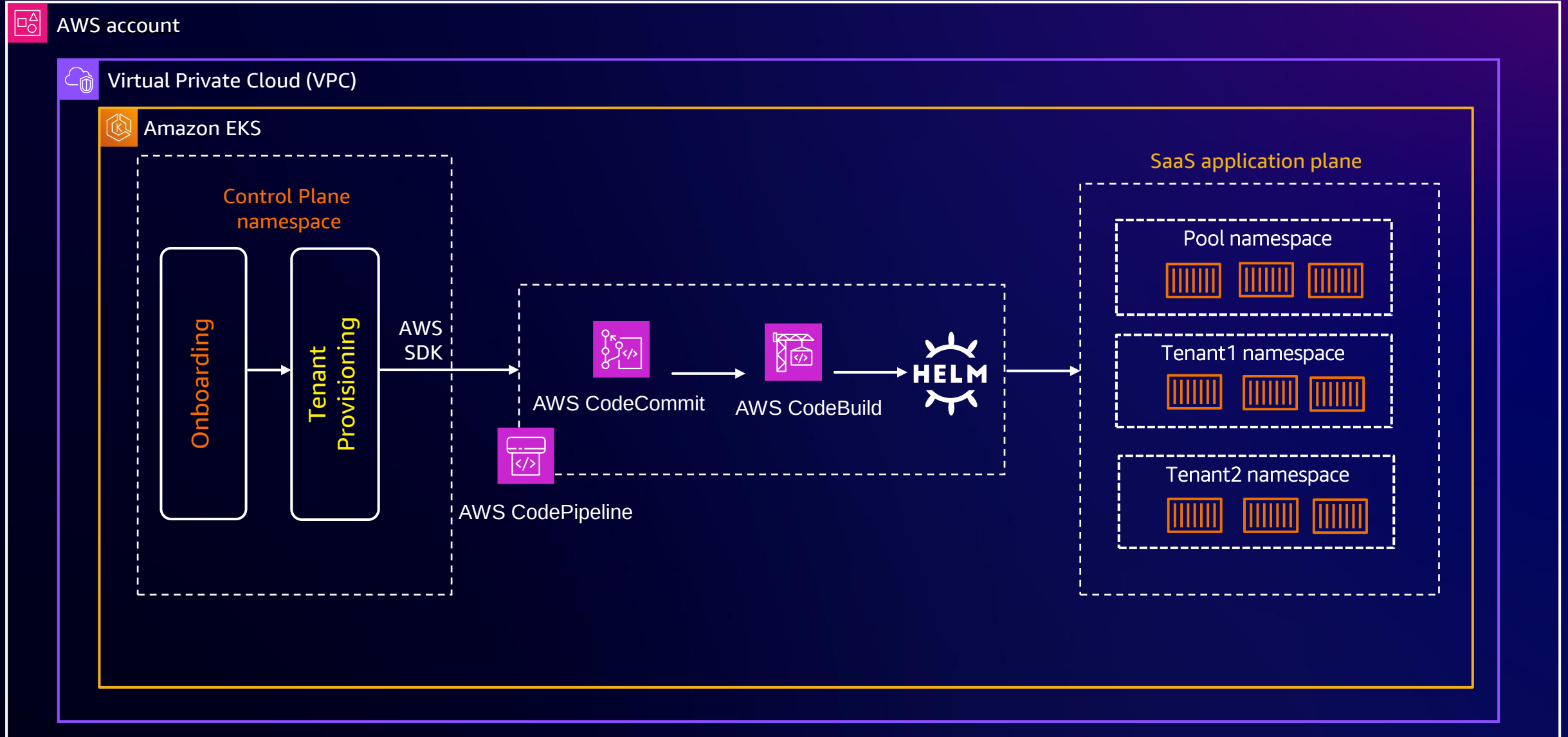
Cloud to on premises



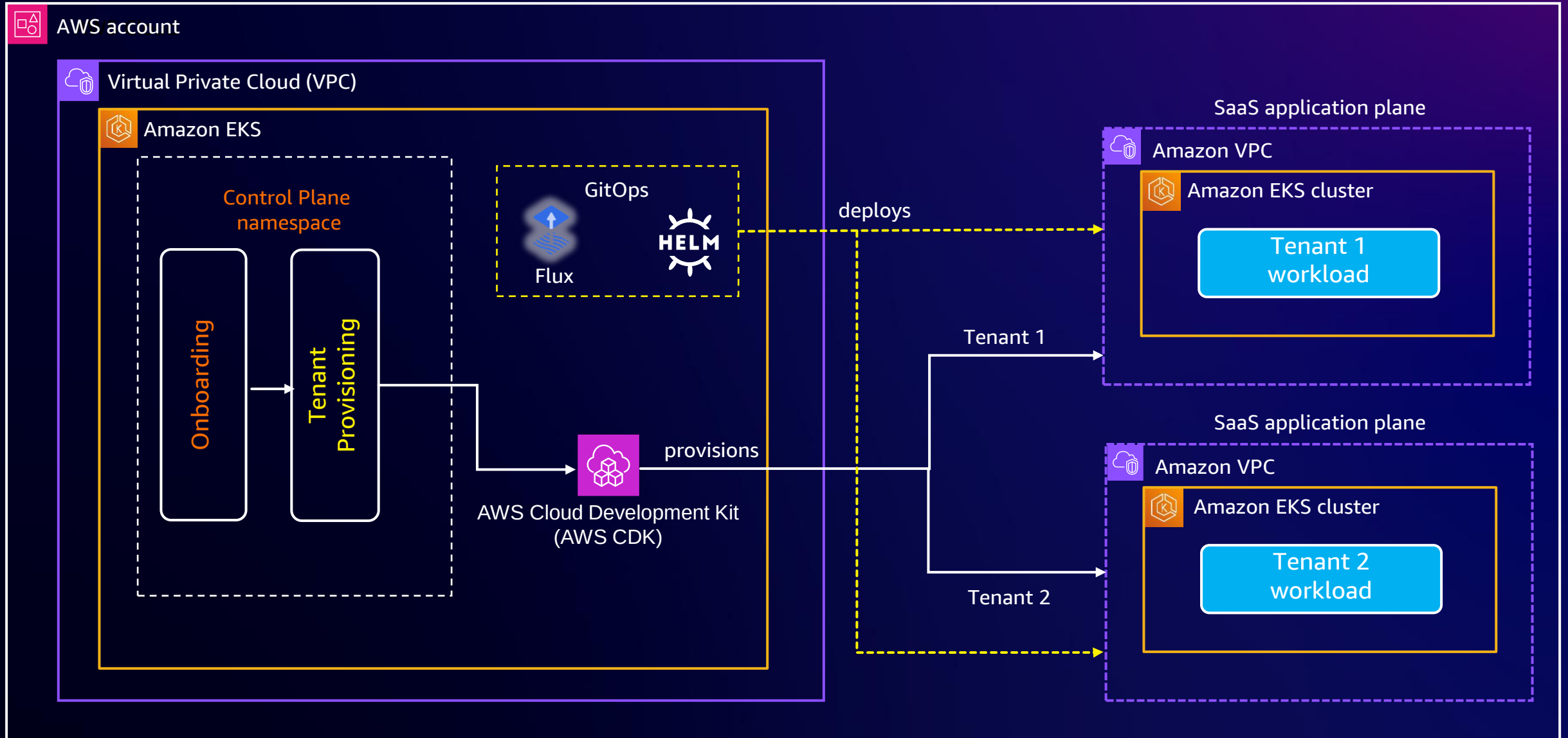
Using a shared AWS account



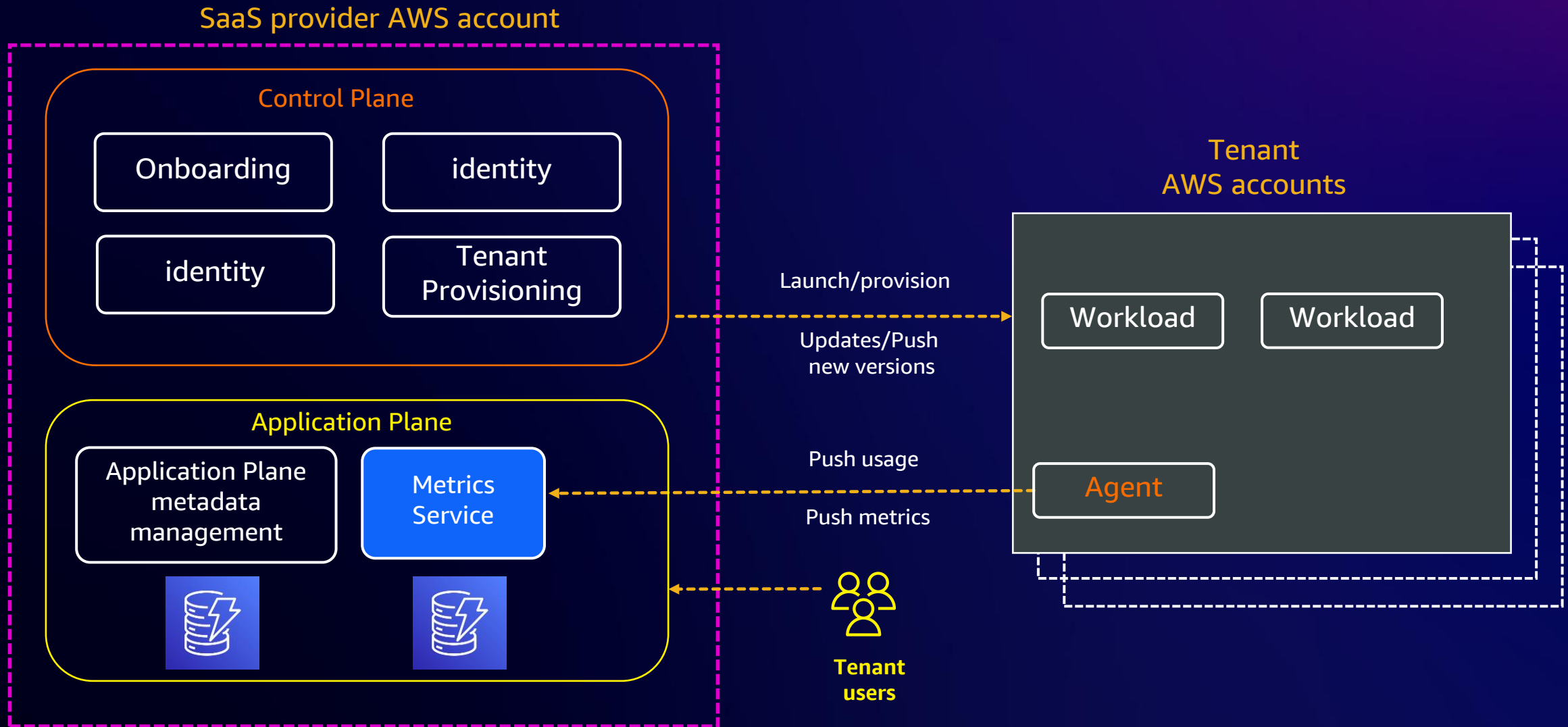
Example: Shared AWS account, shared VPC



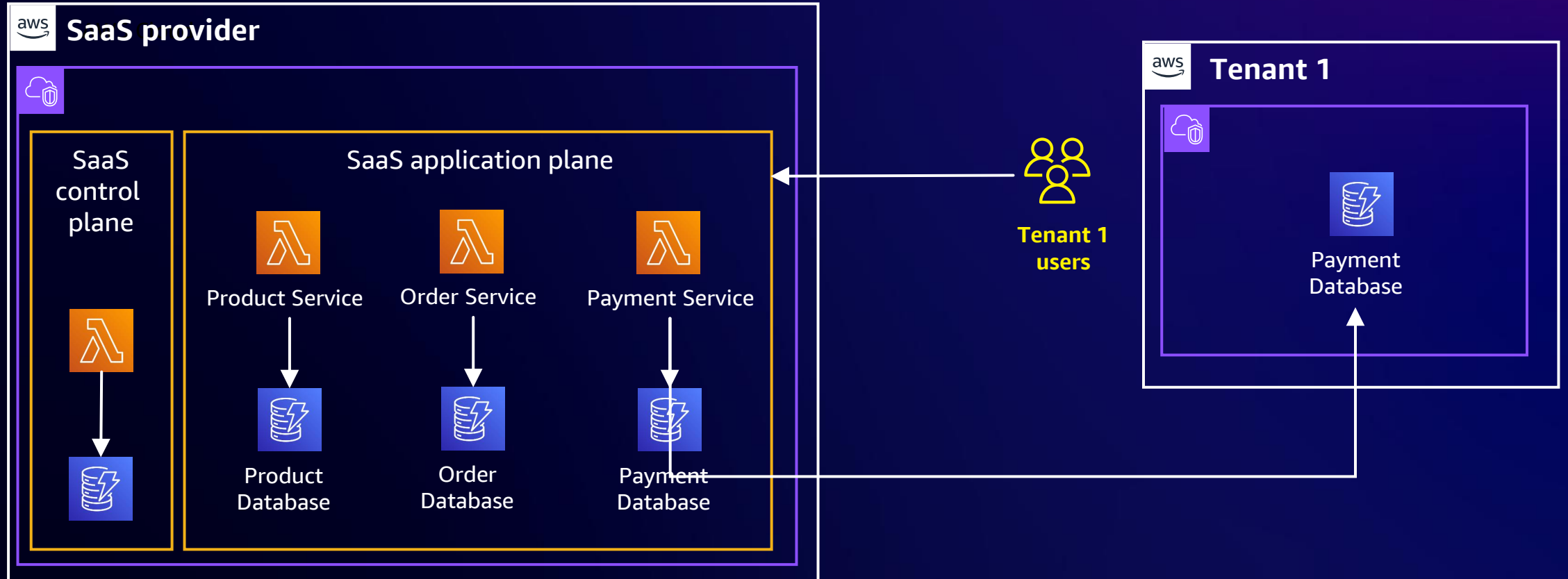
Example: Shared AWS account, separate VPC



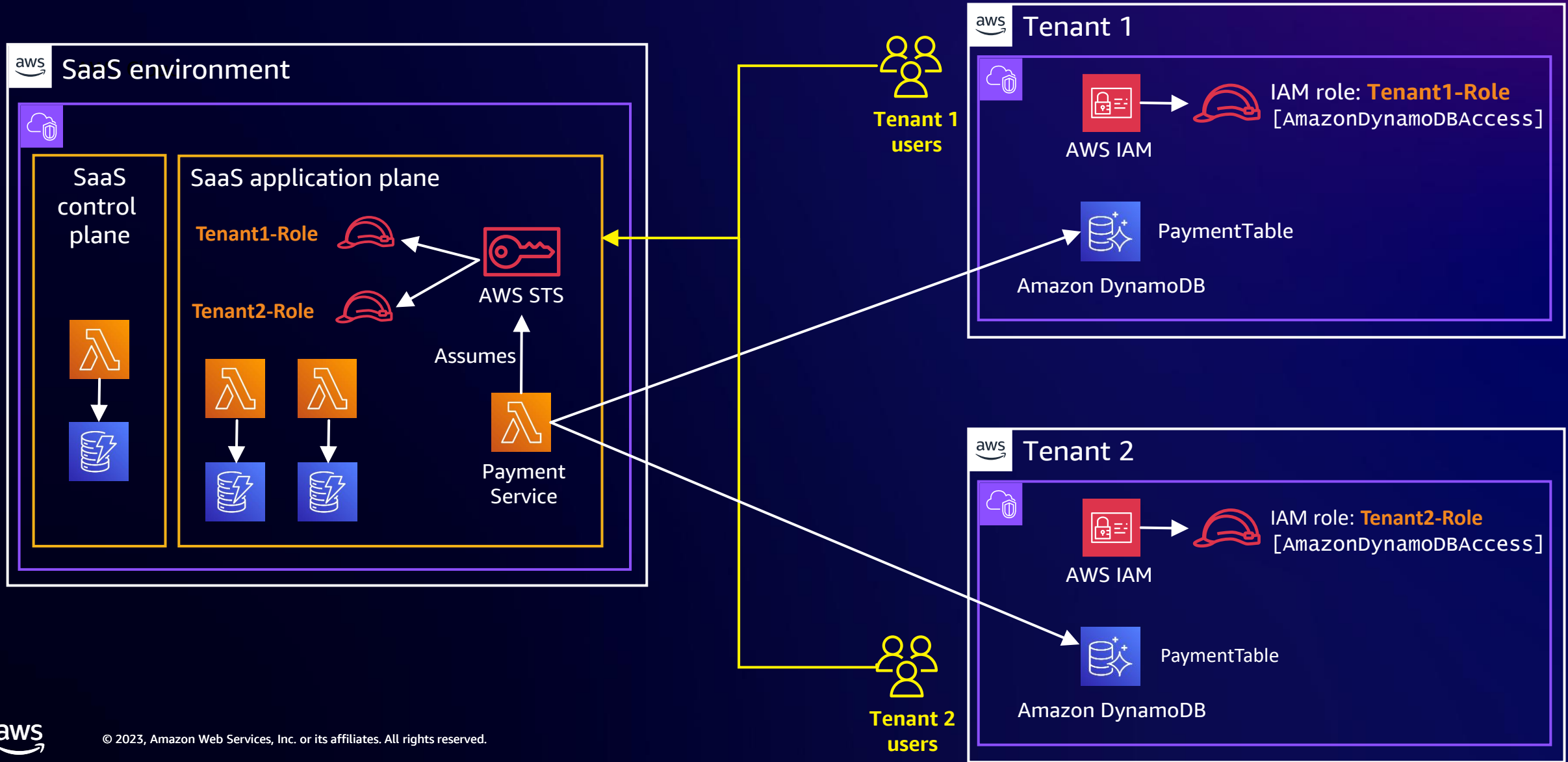
Distributed application planes



Example: E-commerce SaaS



Remote Amazon DynamoDB tables



Tenants provide IAM role with DynamoDB access



Tenant 1 Role ARN : `arn:aws:iam::TENANT1_ACCOUNT:role/Tenant1-Role`

```
{
  "version": "2012-10-17",
  "statement": [
    {
      "action": [
        "dynamodb:GetItem",
        "dynamodb:Query",
        "dynamodb:Scan",
        "dynamodb:BatchGetItem"
      ],
      "resource": "arn:aws:dynamodb:eu-west-1:TENANT1_ACCOUNT:table/PaymentTable",
      "effect": "Allow"
    }
  ]
}
```

Adding a trust policy to tenant IAM role



Tenant1-Role

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::SAAS_PROVIDER_ACCOUNT:role/PaymentService-Execution-Role"
      },
      "Action": "sts:AssumeRole"
    }
  ]
}
```

Payment service needs permission to assumeRole



PaymentService-Execution-Role

```
{  
  "Version": "2012-10-17",  
  "Statement": [  
    {  
      "Effect": "Allow",  
      "Action": "sts:AssumeRole",  
      "Resource": "arn:aws:iam:: TENANT1_ACCOUNT:role/Tenant1-Role"  
    }  
  ]  
}
```



Payment Service



PaymentService
Execution Role

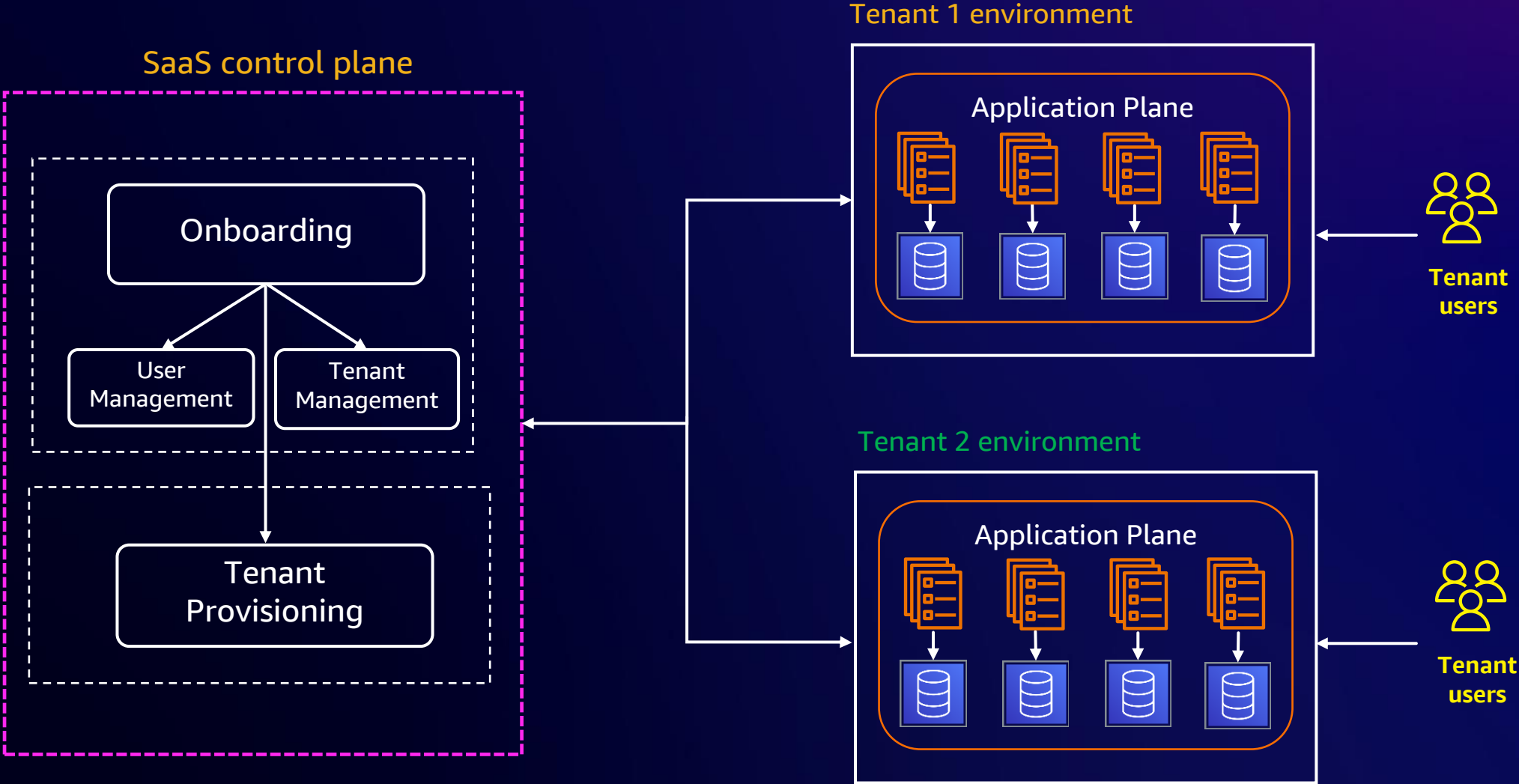


Tenant 1 assumeRole POLICY

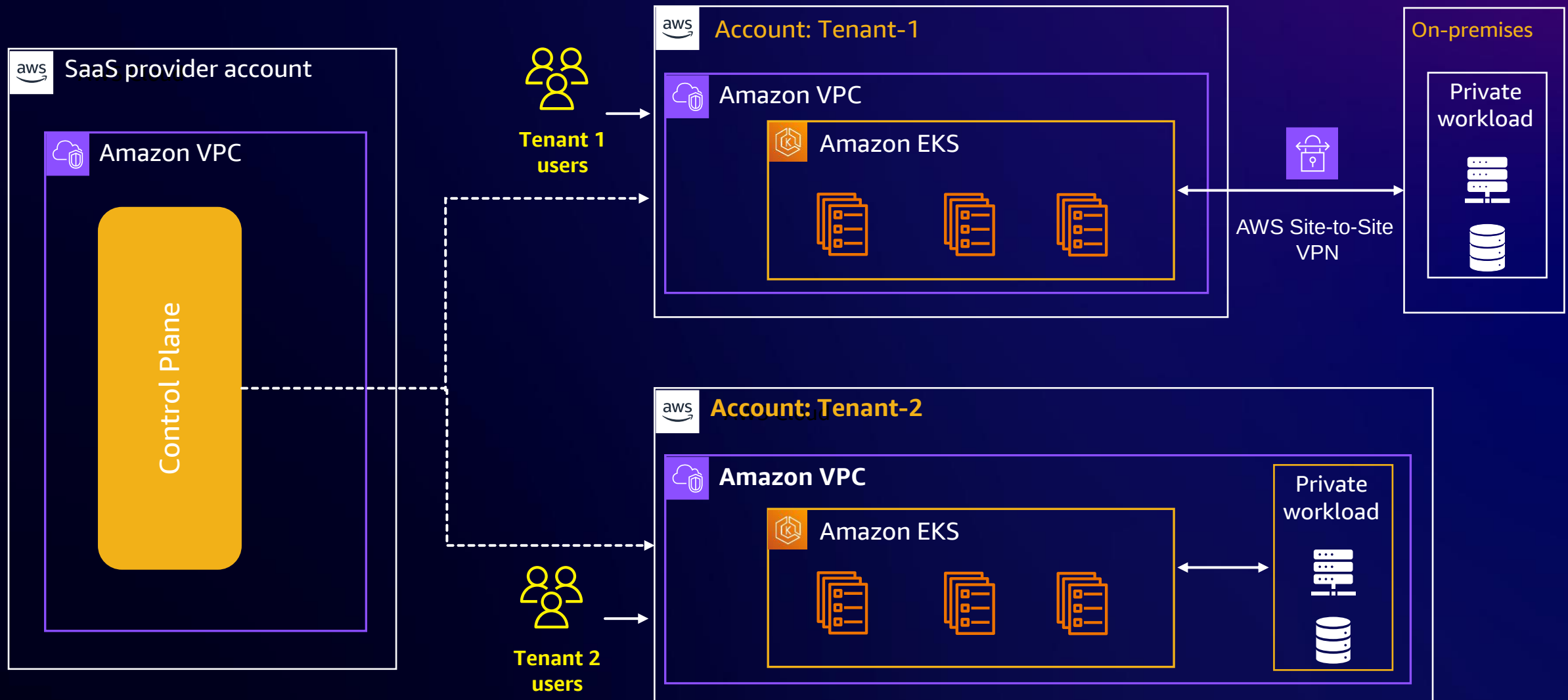


Tenant 2 assumeRole POLICY

Remote application planes



Example: Core banking SaaS



Metrics and talking to the control plane



Key takeaways and suggestions

- When you start matters!
- Don't make it unnecessarily difficult
 - This can be an **incremental** process
 - Build a mental model of your operations
 - Map this model to the services in your control plane
- **Start small** with the **most important** services
 - Identity
 - Onboarding
 - Simple metrics, analytics and system health

Thank you!

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